

27/8/2026

* One-way classified data

Ex: Drug development, different drugs for reducing cholesterol levels.

Sps. there are r diff. types of drugs and n subjects (patients) in the study.

A balanced design would involve applying each treatment (drug type) to the same number of patients so that $mr = n$, where $m = \#$ of replicates.

More generally, treatments will be applied to m_i different subjects (experimental units).

Assignments of treatments to the subjects is done at random (randomization principle).

Response: Y_{ij} = amt. of reduction in cholesterol after one month for the j^{th} subject receiving the i^{th} drug.

i = treatment index $1 \leq i \leq r$

j = replicate index $1 \leq j \leq m_i$

Goal: Compare the efficacy of the different drugs in reducing cholesterol level.

Let P_i denote the prob. dist. of reduction in cholesterol levels for patients receiving drug type i .

- As a 1st approximation assume that the population have the same shape, and are only differing by their means.

$$\mu_i = E(Y_{ij}), \quad i=1, \dots, r$$

Sps. we ~~used~~ want to test $H_0: \mu_i = \mu_{i'}$ v/s $H_1: \mu_i \neq \mu_{i'}$

2 sample t-test: Let $\bar{Y}_i = \frac{1}{m_i} \sum_{j=1}^{m_i} Y_{ij}$

Let $\sigma^2 =$ common variance across the populations.

Estimate of σ^2 is the pooled sample variance:

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^r (m_i - 1) s_i^2}{\sum_{i=1}^r (m_i - 1)}, \quad s_i^2 = \frac{1}{m_i - 1} \sum_{j=1}^{m_i} (Y_{ij} - \bar{Y}_i)^2$$

$$\Rightarrow \boxed{\hat{\sigma}^2 = \frac{\sum_{i=1}^r (m_i - 1) s_i^2}{n - r}} \rightarrow \text{unb. est. of } \sigma^2$$

2 sample t-statistic: $T_{ii'} = \frac{\bar{Y}_i - \bar{Y}_{i'}}{\hat{\sigma} \sqrt{\frac{1}{m_i} + \frac{1}{m_{i'}}}}$

$$T_{ii'} = \frac{\frac{1}{\sigma} \cdot N\left(\mu_i - \mu_{i'}, \sigma^2\right)}{\sqrt{\frac{\hat{\sigma}^2}{\sigma^2}}} \chi_{n-r}^2$$

Noncentrality parameter: $\frac{\mu_i - \mu_{i'}}{\sigma \sqrt{\frac{1}{m_i} + \frac{1}{m_{i'}}}}$

→ Representation of a one-way ANOVA model:

$$Y_{ij} = \mu_i + \varepsilon_{ij}, \text{ where } \varepsilon_{ij} \stackrel{\text{indp}}{\sim} (0, \sigma^2)$$

$$i = 1, \dots, r$$

$$j = 1, \dots, m_i$$

- Differential effect of i^{th} treatment is given by $\mu_i - \mu_0$. μ_0 = overall mean.

- Here, one way to define, μ_0 may be:

$$\mu_0 = \frac{1}{r} \sum_{i=1}^r \mu_i$$

- another possibility

$$\mu_0 = E(\bar{Y}_{..}) = E\left(\frac{1}{n} \sum_{i=1}^r \sum_{j=1}^{m_i} Y_{ij}\right)$$

$$= \frac{1}{n} \sum_{i=1}^r m_i \mu_i$$

- Generally, we may define:

$$\mu_0 = \sum_{i=1}^r w_i \mu_i, \text{ where } \sum_{i=1}^r w_i = 1, 1 > w_i > 0 \forall i.$$

- $\therefore Y_{ij} = \mu_0 + \alpha_i + \varepsilon_{ij}$ where $\alpha_i = \mu_i - \mu_0$ -- (i)
 $i = 1, \dots, r, j = 1, \dots, m_i$

Parameters: $\mu_0, \alpha_1, \dots, \alpha_r, \sigma^2$

α_i = differential effect of i^{th} treatment.

- In (i), we have an identifiability issue

$$Y_{ij} = (\mu_0 - c) + (\alpha_i + c) + \varepsilon_{ij}.$$

To solve this issue:

$$\mu_0 = \sum_{i=1}^r w_i \mu_i = \sum_{i=1}^r w_i (\alpha_i + \mu_0) = \mu_0 \sum_{i=1}^r w_i + \sum_{i=1}^r w_i \alpha_i$$

$$\Leftrightarrow \mu_0 = \mu_0 + \sum_{i=1}^r w_i \alpha_i$$

$$\Leftrightarrow \sum_{i=1}^r w_i \alpha_i = 0 \quad \text{where} \quad \sum_{i=1}^r w_i = 1, \quad 0 \leq w_i \leq 1$$

Note: Natural choices for identifiability constraints are $\sum_{i=1}^r \alpha_i = 0$ or $\sum_{i=1}^r m_i \alpha_i = 0$

- Matrix representation:

$$\begin{bmatrix} Y_{11} \\ Y_{12} \\ \vdots \\ Y_{1m_1} \\ Y_{21} \\ \vdots \\ Y_{2m_2} \\ \vdots \\ Y_{r1} \\ \vdots \\ Y_{rm_r} \end{bmatrix}_{n \times 1} = \begin{bmatrix} 1 & 1 & 0 & 0 & \dots & 0 \\ 1 & 1 & 0 & 0 & \dots & 0 \\ \vdots & 1 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & 1 & 0 & 0 & \dots & 0 \\ \vdots & 0 & 1 & 0 & \dots & 0 \\ \vdots & 0 & 0 & 1 & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & 1 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 0 & 0 & \dots & 0 & 1 \end{bmatrix}_{n \times (r+1)} \begin{bmatrix} \mu_0 \\ \alpha_i \\ \alpha_2 \\ \vdots \\ \alpha_r \end{bmatrix} + \begin{bmatrix} \epsilon_{11} \\ \epsilon_{12} \\ \vdots \\ \epsilon_{1m_1} \\ \epsilon_{21} \\ \vdots \\ \epsilon_{2m_2} \\ \vdots \\ \epsilon_{r1} \\ \vdots \\ \epsilon_{rm_r} \end{bmatrix}$$

$$\therefore X = \begin{bmatrix} 1_{m_1} & 1_{m_1} & 0 & \dots & 0 \\ 1_{m_2} & 0 & 1_{m_2} & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1_{m_r} & 0 & 0 & \dots & 1_{m_r} \end{bmatrix}_{n \times (r+1)}$$

$\downarrow$ $X^{(1)}$ $\downarrow$ $X^{(2)}$ $\dots$ $\downarrow$ $X^{(r)}$

$P(X) = r$
 $2_K = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}_{K \times 1}$

$$\therefore Y = \mu_0 \mathbf{1}_n + \sum_{i=1}^r \alpha_i X^{(i)} + \varepsilon$$

$$\text{Let } \sum_{i=1}^r m_i \alpha_i = 0 \Rightarrow \alpha_r = - \frac{\sum_{i=1}^{r-1} m_i \alpha_i}{m_r}$$

$$\therefore Y = \mu_0 \mathbf{1}_n + \sum_{i=1}^{r-1} \alpha_i \left(X^{(i)} - \frac{m_i}{m_r} X^{(r)} \right) + \varepsilon$$

$$\tilde{X} = [\mathbf{1}_n : \tilde{X}^{(1)} : \dots : \tilde{X}^{(r-1)}], \text{ where } \tilde{X}^{(i)} = X^{(i)} - \frac{m_i}{m_r} X^{(r)}$$

$$\text{Here, } \tilde{X}^{(i)} = X^{(i)} - \frac{m_i}{m_r} X^{(r)}$$

Observation $\mathbf{1}_n$ is orthogonal to all $\tilde{X}^{(i)}$

$$\text{Let } \tilde{X} = [\tilde{X}^{(1)} : \dots : \tilde{X}^{(r-1)}]$$

Q) What are the BLUE's for $\mu_0, \alpha_1, \dots, \alpha_r$ under specific identifiability constraints $\sum w_i \alpha_i = 0$ where $\sum w_i = 1$ & $w_i \geq 0$.

A)	$\mu_0 = \sum_{i=1}^r w_i \mu_i$ $\alpha_i = \mu_i - \mu_0$	$Y_{ij} = \mu_i + \varepsilon_{ij}$ $LSE = \underset{\mu_1, \dots, \mu_r}{\operatorname{argmin}} \sum_{i=1}^r \sum_{j=1}^{m_i} (Y_{ij} - \mu_i)^2$ $\Rightarrow (\hat{\mu}_1, \dots, \hat{\mu}_r) = (\bar{Y}_1, \dots, \bar{Y}_r)$
----	--	--

By G-M theorem BLUE for $\mu_0, \alpha_1, \dots, \alpha_r$ are: $\hat{\mu}_0 = \sum_{i=1}^r w_i \bar{Y}_i$, $\hat{\alpha}_i = \bar{Y}_i - \hat{\mu}_0$ for $i=1, \dots, r$.